

ITA0608 – Machine Learning Lab Practicals

EXP 08:

Write a program to implement Linear Regression (LR) algorithm in python

Aim:

To implement the Linear Regression (LR) algorithm in Python and classify data using an appropriate dataset.

Algorithm:

- Load the dataset and split it into training and testing datasets.
- Perform feature scaling to normalize the input data.
- Train the Linear Regression model using the training data.
- Predict the class labels for the testing data.
- Evaluate the model using the Confusion Matrix, Accuracy, and classify a new sample.

Program:

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

diabetes = load_diabetes()
X = diabetes.data
y = diabetes.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("Mean Squared Error (MSE):", round(mse, 2))
print("R2 Score:", round(r2, 2))
new_sample = [X[0]]
```

```
prediction = model.predict(new_sample)
print("\nPredicted Value for New Sample:", round(prediction[0], 2))
```

Output:

```
... Mean Squared Error (MSE): 2821.75
    R2 Score: 0.48

    Predicted Value for New Sample: 209.69
```

Result:

The Linear Regression (LR) algorithm was successfully implemented in Python using the Iris dataset.